Laguna State Polytechnic University – College of Computer Studies
ITEC 102 – Fundamentals of Programming

User Manual: Interactive C# Console-Based Tic-Tac-Toe Game Application

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1. System Overview
The Interactive C# Console-Based Tic-Tac-Toe Game Application is a lightweight, two-player console game built in C#. It replaces manual pen-and-paper tracking with an automated 3×3 grid, single-keystroke controls, real-time move validation, multi-round score tracking, and a final scoreboard summary.

2. System Requirements
• Device: Any computer or mobile device with a modern web browser or .NET runtime.
• Environment: OnlineGDB (C# Compiler), Visual Studio, or the .NET SDK.

3. How to Run the Application
3.1. Download or copy the source code from src/Program.cs.
3.2. Open OnlineGDB (onlinegdb.com), select C# as the language, and paste the code (or open Program.cs in Visual Studio / .NET CLI).
3.3. Click Run (or type dotnet run) to launch the console interface.

4. Step-by-Step Usage Instructions
Step 1: Register Player Names
• When the program launches, the TIC TAC TOE PROJECT banner appears.
• Enter a name for Player 1 (assigned symbol [X]) and press Enter.
• Enter a name for Player 2 (assigned symbol [O]) and press Enter.
• Note on Name Length: Player names are limited to a maximum of 10 characters to preserve the alignment of the console UI borders. If a name exceeds 10 characters (for example, testName101), the system automatically trims it to the first 10 characters (testName10).
Step 2: Play the Match
• The game displays the ITEC 102: TIC-TAC-TOE header, a numbered 1–9 board on the left, and the MATCH METRICS scoreboard on the right.
• On your turn, press a single numeric key from 1 to 9 (Console.ReadKey) corresponding to an open cell on the grid—no Enter key is needed.
• If you press an invalid key or choose an already occupied slot, the system rejects the move and prompts you to choose again.
• Turns alternate automatically until a player aligns three symbols horizontally, vertically, or diagonally, or until all 9 cells are filled (resulting in a draw).

Step 3: Round End & Restart Options

• At the end of a round, the winner (or draw status) is announced and the MATCH METRICS score updates automatically.

• Use a single keystroke at the bottom prompt:

    • Press 1 to End Game and proceed to the final summary.

    • Press 2 to Restart Game and clear the board for another round while keeping the running score.

Step 4: View Final Scores & Exit

• Pressing 1 opens the FINAL SCORES screen displaying total round wins for both players and a closing thank-you message.

• Press Enter to exit the console safely.

5. Sample Inputs & Expected Outputs

Game Stage	Sample Input	Expected Console Output
Name Registration	testName101(Player 1) name(Player 2)	Trims player 1's name to testName10 (10-char limit), saves name, and loads the board with Score: P1 (0) vs P2 (0).
Board Move	Press 1 (single key)	Place X in slot 1 immediately and switches the turn promt to Player 2.
Round Victory	Press 7 (completes column 1-4-7)	Displays testName10 WINS THIS ROUND!, updates score to P1 (1) vs P2 (0), and shows [1] End Game | [2] Restart Game.
End Session	Press 1 (single key)	Displays the FINAL SCORES banner (testName10 (1) | (0) name) and terminates cleanly.